DESIGN AN IMPLEMENTATION OF IRS FOR SMALL MIMO WIRELESS NETWORK

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The **exponential growth of wireless data demand** and the **limitations of conventional communication infrastructure** have necessitated the development of **innovative technologies** to improve **signal quality** and **energy efficiency**. **Intelligent Reflecting Surfaces (IRS)** have emerged as a **promising solution** in **next-generation wireless networks**.

This project explores the **design and implementation** of an **IRS-assisted wireless communication system**, where a **programmable meta surface** is used to **dynamically control** the reflection of incident signals toward the receiver, thereby **enhancing the overall communication performance**.

The **IRS is modelled using discrete phase shifts** and is integrated into a **simulation environment** to analyse its impact on system parameters such as **signal-to-noise ratio (SNR)**, **bit error rate (BER)**, and **achievable data rate**. **Optimization algorithms** are employed to configure the IRS elements in real-time, **maximizing signal strength** at the receiver while **minimizing interference**.

The project demonstrates that **IRS can significantly improve wireless coverage, reliability, and spectral efficiency**, especially in **non-line-of-sight (NLoS) scenarios**.

This work contributes toward the **practical realization of IRS in 6G networks**, highlighting its potential in **future smart radio environments**.

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ABBREVIATIONS

S. NO.	ABBREVIATION	EXPANSION
1	IRS	**I**ntelligent **R**eflecting **S**urface
2	MIMO	**M**ultiple **I**nput **M**ultiple **O**utput
3	IoT	**I**nternet **O**f **T**hings
4	NLOS	**N**on-**L**ine-**O**f-**S**ight
5	SNR	**S**ignal-to-**N**oise **R**atio
6	FPGA	**F**ield-**P**rogrammable **G**ate **A**rray
7	RF	**R**adio **F**requency
8	WSN	**W**ireless **S**ensor **N**etwork
9	CSI	**C**hannel **S**tate **I**nformation
10	AI	**A**rtificial **I**ntelligence
11	ML	**M**achine **L**earning
12	WPCN	**W**ireless **P**owered **C**ommunication **N**etwork
13	ISWPT	**I**ntegrated **S**ensing and **W**ireless **P**ower **T**ransfer
14	SDP	**S**emi-**D**efinite **P**rogramming
15	LC	**L**ow **C**omplexity
16	MM	**M**ajorization-**M**inimization
17	HAP	**H**igh-**A**ltitude **P**latform
18	WD	**W**ireless **D**evices
19	AR/VR	**A**ugmented **R**eality / **V**irtual **R**eality
20	FEA	**F**inite **E**lement **A**nalysis
21	GUI	**G**raphical **U**ser **I**nterface
22	EMC	**E**lectro-**M**agnetic **C**ompatibility
23	MEMS	**M**icro-**E**lectro-**M**echanical **S**ystems
24	SISO	**S**ingle **I**nput **S**ingle **O**utput
25	OFDM	**O**rthogonal **F**requency **D**ivision **M**ultiplexing
26	URLLC	**U**ltra-**R**eliable **L**ow-**L**atency **C**ommunication

27	Wi-Fi	**W**ireless **F**idelity
28	AM	**A**mplitude **M**odulation
29	FM	**F**requency **M**odulation
30	PSK	**P**hase **S**hift **K**eying
31	FSK	**F**requency **S**hift **K**eying
32	APES	**A**mplitude and **P**hase **E**stimation